

ECN 711
Problem Set #5

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Due date: October 5th, 2025.

1. Consider the overlapping generations model with capital. Time is discrete ($t = 0, 1, 2, \dots$). Individuals live for two periods, and only work when young.

An individual born at time t chooses consumption, savings and hours worked when young to maximize:

$$U(c_{y,t}, c_{o,t+1}) = \log(c_{y,t}) + \beta \log(c_{o,t+1})$$

with $\beta > 0$.

Competitive firms have access to a production technology that uses capital and labor services, and satisfies all standard assumptions. The size of new cohorts grows at the (gross) rate $1 + n$, and labor efficiency grows at the rate $1 + g$.

1. Derive the law of motion for the capital to effective labor ratio. Argue that along a balanced-growth path, (i) the investment to output ratio is constant, and (ii) consumption of old and young individuals grows at the (gross) rate $1 + g$.
2. Suppose at time $t = t_0 - 1$ the economy is in the steady state. At the start of t_0 , the government of *Amerika* decides to provide aid to this economy, in the fixed amount of x per young person, once and for all.

Evaluate the following statement: *As the result of foreign aid, output per worker will be higher in the long run than in the absence of foreign aid. Furthermore, the higher the transfer, the higher the impact on long-run growth in output per worker.* Explain.

2. Consider the following version of the overlapping generations model with capital. Time is discrete ($t = 0, 1, 2, \dots$). Individuals live for two periods, and work in both periods of their lives. Individuals are identical within a generation, and are endowed with labor efficiency units in each period that they devote entirely to market work. In the first period, their labor efficiency is ω_y , while in the second period their labor efficiency is ω_o .

An individual born at t chooses consumption levels $c_{y,t}$ and $c_{o,t+1}$ to maximize

$$\log(c_{y,t}) + \beta \log(c_{o,t+1})$$

There is a large number of individuals in each generation, and the size of each generation is normalized to 1. Competitive firms have access to a Cobb-Douglas technology that uses capital and labor. Capital fully depreciates during a period.

1. Formulate the problem of a young agent in period t .
2. Obtain the equilibrium law of motion for the capital-labor ratio. Is the steady state for this economy unique?
3. Suppose now that for reasons exogenous to this problem, the government permanently restricts the hours that old individuals can work. This event occurs unexpectedly at $t = t_0$, and the effective labor endowment for old individuals becomes $\omega_o \lambda$, $\lambda \in (0, 1)$.

Describe the consequences over time of this policy on prices. How does the capital to labor ratio compare across steady states? Explain.

3. Consider the following version of the overlapping generations model with capital and government consumption. Time is discrete ($t = 0, 1, 2, \dots$). Individuals are identical within a cohort, live for two periods and work only when young.

An individual born at t chooses consumption levels $c_{y,t}$ and $c_{o,t+1}$ to maximize

$$\log(c_{y,t}) + \beta \log(c_{o,t+1})$$

Competitive firms have access to a Cobb-Douglas technology that uses capital and labor. Suppose for simplicity that growth in population and labor efficiency are zero.

There is a government in this economy that consumes the aggregate amount G_t in each date. Government consumption is financed by a tax on the (net) return on savings of old individuals at the rate $\tau_t \in (0, 1)$.

1. Formulate the problem of a young agent in period t . Derive the first-order conditions that characterize its solution.
2. Define a competitive equilibrium.
3. Evaluate the following statement. ‘Consider two economies with the same preferences and technology as above, and the same initial conditions. Government consumption in economy A is always at least the same as in economy B (i.e. $G_{A,t} \geq G_{B,t}$, for all t). Therefore, interest rates in A will always be higher or equal than in B.’ Explain carefully.
4. Consider the following version of the overlapping generations model with capital and government consumption. Time is discrete ($t = 0, 1, 2, \dots$). Individuals are identical within a cohort, live for two periods and work only when young.

An individual born at t chooses consumption levels $c_{y,t}$ and $c_{o,t+1}$ to maximize

$$v(c_{y,t}) + \beta v(c_{o,t+1}),$$

where v is utility function with elasticity of intertemporal substitution in consumption $1/\sigma$, $\sigma \neq 1$. Competitive firms have access to a Cobb-Douglas technology that uses capital and labor, with capital share α . There is full depreciation (i.e. $\delta = 1$).

Let Z be the TFP associated to the production technology. Your job is to determine some of the steady-state consequences of changes in Z .

1. Derive the equilibrium law of motion for the capital to effective labor ratio.
2. Let ϕ denote the capital to output ratio. Show that the steady-state value for ϕ is independent of the level of TFP. Show that the rate of return on capital is independent of the level of TFP. Explain carefully.
3. Show that there is a unique steady-state value for ϕ , and therefore a unique steady-state value for the capital to effective labor ratio.
5. (Bonus Exercise) This question is about the notion(s) of GDP in aggregate economies, so you are about to do engage in some serious national income accounting.

There are three (equivalent) notions of GDP. The first one, GDP_1 , states that GDP is the market value of all final sales of goods and services. The second one, GDP_2 , states that GDP is the value of all payments to factors of production. The third one, GDP_3 , states that GDP is the total value added in all sectors of the economy, where valued added in each sector is sales net of intermediate goods.

Consider an economy comprised of $N + 1$ sectors. Sectors $j = 1, \dots, N$ produce intermediate goods, that are used in the production of other intermediate goods $i \neq j$ and in the production of the final good. Sector $N + 1$ produces a single final good.

For simplicity, we assume that there is a single firm in each sector. Final goods are produced according to a technology that uses goods from all sec-

tors. That is, $Y_{N+1} = F(y_1, y_2, \dots, y_N)$, where y_j stands for the sales of each sector j to the final producer firm.

Each sector j , $1 < j \leq N$ produces according to a technology that uses capital (k), labor (l), and the purchases of goods from sector $j - 1$, x_{j-1} . Hence, production of intermediate good j , is given by

$$z_j = G_j(k_j, l_j, x_{j-1})$$

Finally, sector $j = 1$ produces according to a technology that only uses capital and labor. Therefore,

$$z_1 = G_1(k_1, l_1)$$

Let p_j be the price of each good in the economy, $j = 1, \dots, N + 1$. Let W and R the rental prices for labor and capital services.

1. Define GDP_1 .
2. Show that $GDP_2 = GDP_3$ (Hint: Define profits in each sector and aggregate across all sectors. Note that production of each intermediate good is divided between sales to the final good firm and sales to other intermediate goods producers).
3. Show that $GDP_1 = GDP_2$, and therefore argue that the three definitions are equivalent.